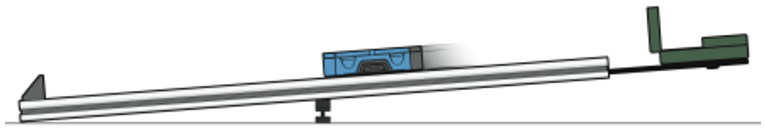
Name: Date: Period:

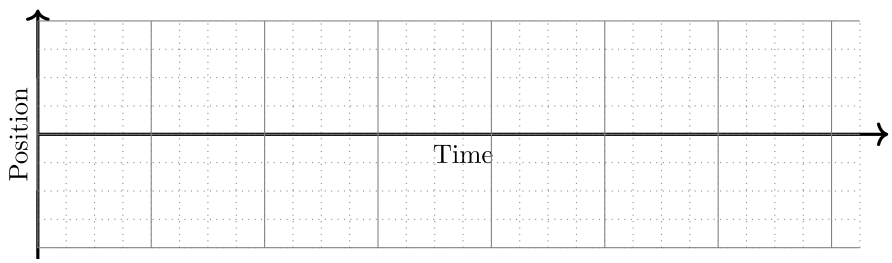
# Motion Graph Stations

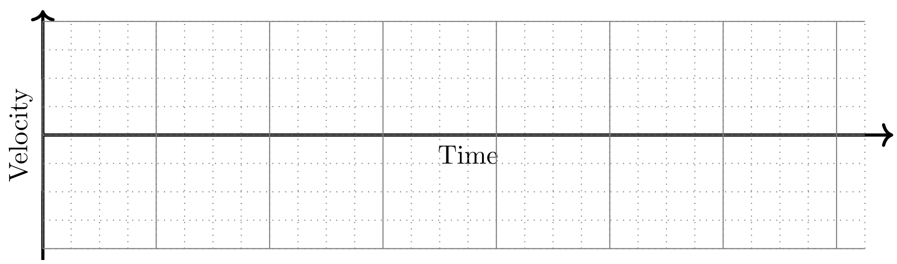
*Stations 1-2 are based off of Experiment #2 in* Physics with Vernier*, and the images have been taken from there.*

## 1 Cart on an Incline



1. Place the cart on the track near the bottom. Press the play button to start data collection and then give the cart a push up the incline. Let the cart roll freely up nearly to the top, and then back down. Keep your hands away from the track as the cart rolls, then catch the cart as it nears the end stop. The cart should not get closer than 0.15 m to the motion detector. If you do not see a smooth graph, the cart was most likely not in the beam of the motion detector. Make adjustments and repeat data collection until you get a smooth graph.
2. Sketch the position and velocity graphs below. Please **be careful** with the horizontal axis. Try to make it so that each graph has the same events occuring at the same location along the $t$-axis. Don't rush this step.



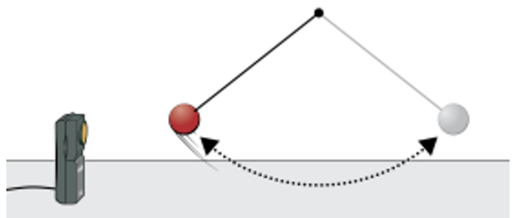




3. During which interval of time in the motion was the acceleration constant? How can you tell?

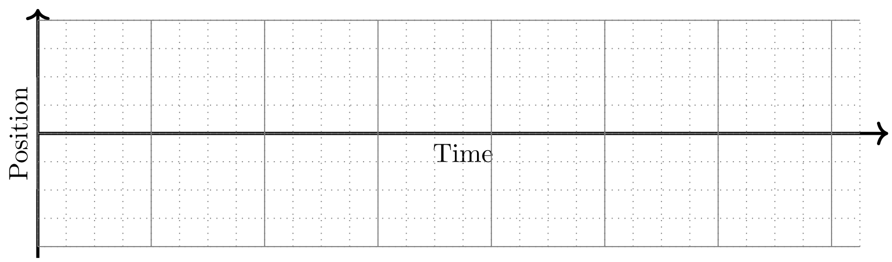
4. Was there any point in the motion where the velocity was zero? Explain.

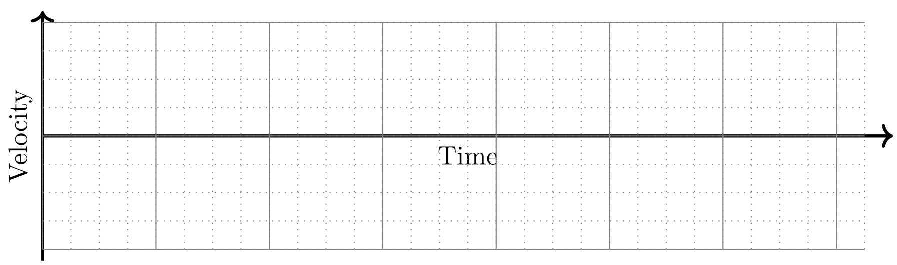
5. What is interesting about the *velocity graph* at the point in time when the cart is at its largest position?

# 2 Pendulum



1. Place the motion detector near the pendulum. The motion detector should be level with, and about 1 m away from, the pendulum bob when it hangs at rest. The bob should never be closer to the detector than 25 cm. Pull the pendulum about 15 cm toward the motion detector and release it to start the pendulum swinging.
2. Press play to start data collection.
3. When data collection is complete, a graph of position vs. time is displayed. If you do not see a smooth graph, the pendulum was most likely not in the beam of the motion detector. Make adjustments and repeat until you get a smooth graph.
4. Sketch the position and velocity graphs below. Please **be careful** with the horizontal axis. Try to make it so that each graph has the same events occuring at the same location along the $t$-axis. Don't rush this step.



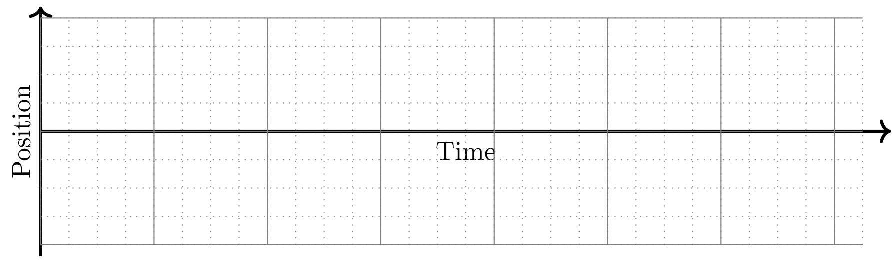

5. Was the acceleration constant or changing? How can you tell?
6. Was there any point in the motion where the velocity was zero? Explain.
7. Look carefully at your graphs. When the position graph is at a maximum or minimum, what is true about the velocity graph? Why does this make sense?

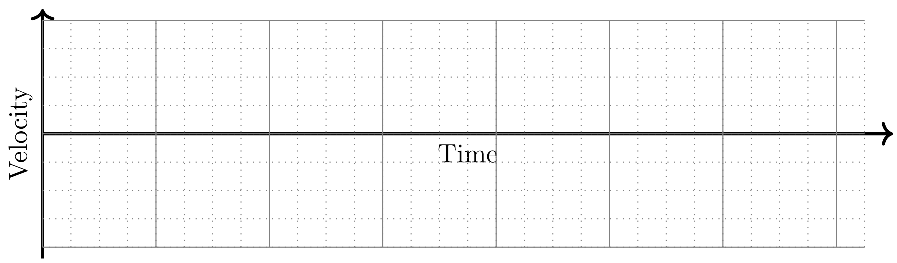
Name: Date: Period:

# 3 Bouncing Basketball

*Credit to Dr. Dennis Krause, Wabash College, for the inspiration behind Station #3.*

1. Hold the basketball about 50 cm away from the motion detector. Start taking data, and then drop the ball, allowing it to bounce a few times.
2. See if your data looks smooth. If there have been issues, take a new data set before proceeding.
3. Sketch the position and velocity graphs below. Please **be careful** with the horizontal axis. Try to make it so that each graph has the same events occuring at the same location along the $t$-axis. Don't rush this step.




4. Why were there "breaks" in the velocity graph where the data jumped from negative to positive? What happened at these locations?

5. Were there any points in the motion where the velocity was zero? Explain.

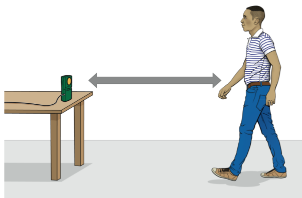
6. Was there any intervals of time in the motion where the acceleration was constant? Explain.

Name: Date: Period:

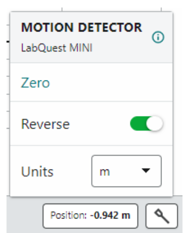
# 4 Graph Match

*This station is based off of Experiment #1 in* Physics with Vernier*, and the images have been taken from there.*

## Setup



1. Connect the motion detector to your computer. Launch `Graphical Analysis`.
2. Place the motion detector so that it points toward an open space at least 2 m long. Use short strips of masking tape on the floor to mark distances of 0.5 m, 1 m, 1.5 m, and 2 m from the motion detector.
3. Find the position on the ground marked "0 meters" and stand there. Have a lab partner click the "Position" meter at the bottom right corner of the screen, "Reverse" the sensor, and click "Zero."




4. Monitor the position readings and practice walking toward the motion detector. Sometimes you will have to walk backwards. Confirm that the position values make sense as you move back and forth.

## Position Graph Matching

5. Click or tap Graph Tools, , and choose Add Graph Match. Choose Position. A position target graph is displayed for you to match.
6. Think about how you would walk to reproduce the target graph. To test your prediction, choose a starting position. Start data collection, then walk in such a way that the graph of your motion matches the target graph on the screen.
7. If you were not successful, start data collection again when you are ready to begin walking. Repeat this process until your motion closely matches the graph on the screen.
8. Have every member of your group try. To perform a second position graph match, click or tap Graph Tools, , choose Add Graph Match, and then Position.

## Velocity Graph Matching

9. `Graphical Analysis` can also generate random target velocity graphs for you to match. Click or tap Graph Tools, , and choose Add Graph Match. Choose Velocity to view a velocity target graph.
10. Try to reproduce the target graph. Keep repeating until you get something that closely matches.
11. Repeat for every member of your group.

## Analysis

1. Explain the significance of the slope of a position graph. Include a discussion of positive and negative slope.

2. What type of motion is occurring when the slope of a position graph is zero?

3. What type of motion is occurring when the slope of a position graph is constant?

4. What type of motion is occurring when the slope of a *velocity* graph is zero?